

DEVELOPING A LIVE 3D TELEMETRY VISUALIZATION FOR STAIRLIFT FAULT DIAGNOSIS

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Abstract. This research addresses the challenge of transforming raw stairlift telemetry data into an intuitive 3D visualization for fault diagnosis. The core technical problem is the live mapping of noisy, asynchronous sensor readings onto a curved 3D rail geometry with segment-dependent encoder-to-distance ratios. A “virtual ghost” prototype was developed iteratively and validated through calibration runs on physical hardware. The results show that a processing pipeline combining non-linear encoder mapping, recursive state estimation, and smooth tangent-based orientation reconstruction produces a stable live 3D representation. This enables faster spatial interpretation of stairlift position and fault conditions compared to text-based error codes.

1 Introduction

DeVi Comfort designs and manufactures stairlift systems and develops the accompanying control software in-house. A stairlift consists of a chair unit that travels along a custom-bent rail mounted to a staircase. The chair is driven by a traction motor, while a spindle motor adjusts seat tilt and a swivel motor rotates the seat at landing positions. During operation, embedded controllers continuously produce telemetry: traction encoder pulses, spindle encoder values, swivel angles, and system status bits including error codes. For service, maintenance, and further development, engineers need to quickly understand the current state of a stairlift: where the chair is positioned on the rail, whether it is moving correctly, and what deviations or faults have occurred.

Currently, fault information from the controller is presented as textual error codes and numeric status values. While technically correct, these require significant domain knowledge to interpret spatially. An error code indicating a position deviation does not convey where on the rail the deviation occurred, in which direction, or how large the discrepancy is.

The underlying technical challenge is twofold. The raw telemetry data (encoder pulse counts, spindle values, swivel angles) must be mapped onto a three-dimensional rail geometry. This rail contains straight sections, vertical bends, and horizontal bends, each consuming a different number of encoder pulses per millimeter of travel. Additionally, the sensor readings arrive at irregular intervals over a wireless serial link with noise, latency, and occasional packet loss. The visualization must bridge the gap between these asynchronous measurements and a continuous, smooth spatial representation that updates at display frame rate.

A live 3D visualization of stairlift position and movement would have direct practical value. Service engineers could visually identify whether the chair has lost its expected position, is stuck in a bend, or exhibits abnormal tilt. This would reduce diagnosis time and minimize unnecessary on-site visits. For the development team,

a live telemetry view would enable rapid verification of firmware behavior during testing and commissioning of new rail configurations.

The goal of this research is to determine which data processing and mathematical models are needed to transform live stairlift telemetry into a reliable and intuitive 3D visualization for fault analysis. The central research question is:

What data processing pipeline and mathematical models are required to transform live stairlift telemetry into a 3D visualization that enables faster and more intuitive fault diagnosis compared to text-based error codes?

The research follows an iterative prototyping approach based on the spiral model [1]. Each iteration consists of analysis, design, implementation, and validation against practical scenarios on physical hardware. The primary methods are formal mathematical analysis, architectural design, and experiment with measurable accuracy criteria.

Section 2 presents the problem statement, structured as a problem analysis, prior work survey, and the resulting research questions. Section 3 describes the research methods per subquestion. Section 4 presents the findings. Section 5 concludes with answers to the research questions and recommendations.

2 Problem Statement

2.1 Problem Analysis

2.1.1 Sensor-to-Spatial Mapping in Mechatronic Systems

The challenge of transforming discrete sensor readings into a continuous spatial representation is well-established in mechatronic system diagnostics [2]. In the broader context of digital twin research, Bofill et al. identify real-time sensor data integration as one of the primary technical barriers to effective fault monitoring [3]. The fundamental difficulty is that sensors produce scalar

values (counts, voltages, status bits) at discrete intervals, while a spatial visualization requires a continuous 3D position and orientation that updates at display frame rate.

In robotics and vehicle tracking, this problem is typically simplified by the assumption that sensor readings map linearly to spatial coordinates [4]. A rotary encoder on a linear actuator, for example, produces pulses proportional to displacement, and the mapping from pulse count to position is a single constant. This assumption breaks down when the trajectory is non-linear: on a curved path, the relationship between encoder pulses and spatial displacement depends on the local curvature [5].

2.1.2 Non-Linear Encoder Mapping on Curved Rails

A stairlift rail is composed of straight sections, vertical bends (pitch changes), and horizontal bends (yaw changes). The mechanical drive system uses a traction motor with an incremental encoder to measure displacement [6]. Due to the geometry of the rail bending mechanism, each segment type consumes a different number of encoder pulses per unit of physical travel. A vertical upward bend, for instance, requires fewer pulses per millimeter of arc length than a downward bend, because the traction wheel follows the inner or outer radius of the curve respectively.

This means that a single scalar encoder reading cannot be converted to a 3D position without a model of the full rail topology. The mapping must account for the cumulative pulse cost of each preceding segment, making it a piecewise non-linear function rather than a simple linear scaling. Without such a model, position errors accumulate at every bend transition, growing proportionally to the number of bends in the rail.

2.1.3 State Estimation from Noisy Telemetry

The telemetry link between the stairlift controller and the visualization application introduces additional challenges. The Bluetooth Classic Serial Port Profile [7] provides a virtual serial connection at approximately 20 Hz, but with variable latency (10–80 ms per round trip) and occasional packet loss. The visualization, however, must render at 60 fps (16.7 ms per frame).

This creates a temporal mismatch: new sensor data arrives roughly every third frame, and frames between measurements must be filled with predicted values. If the most recent measurement is simply held constant until the next arrives, the display exhibits visible staircase-like jumps. If measurements are used directly without filtering, encoder quantization noise (inherent to incremental encoders [5]) causes jitter. Both artifacts undermine the interpretability of the visualization.

The general solution to this class of problem is recursive state estimation: a filter that maintains a probabilistic model of the system state and updates it with each new measurement [8]. The choice of filter, its process model, and its tuning determine the trade-off between responsiveness and smoothness.

2.1.4 Multi-Axis Orientation from Independent Sensors

The stairlift chair has four independent rotational degrees of freedom: yaw (direction along the rail), pitch (rail inclination), seat tilt (spindle motor), and seat rotation (swivel motor). Each is measured by a separate sensor with its own update rate and noise characteristics. Reconstructing the full 3D orientation of the chair requires combining these independent streams into a single consistent rotation, avoiding mathematical singularities that can occur when multiple rotations are composed [9].

2.1.5 Summary of Technical Subproblems

Three interrelated technical subproblems emerge from this analysis:

1. **Non-linear encoder-to-position mapping:** Converting a scalar encoder count into a normalized position along a 3D curve with segment-dependent pulse costs.
2. **State estimation:** Producing a smooth, low-latency estimate of position and velocity from noisy, asynchronous measurements arriving at a lower rate than the display refresh.
3. **Multi-axis orientation reconstruction:** Combining four independent rotational sensor streams into a singularity-free 3D orientation.

2.2 Prior Work

2.2.1 State Estimation

For state estimation from noisy sensor data, three approaches are commonly used in embedded and robotics applications: moving average filters, complementary filters, and Kalman filters.

A **moving average filter** computes the mean of the last N samples and outputs it as the filtered value. Smith shows that for purely time-domain noise reduction on stationary or slowly varying signals, moving averages are close to optimal and have the advantage of computational simplicity [10]. They are therefore a strong choice for applications where the input signal changes slowly relative to the sampling rate and no prediction is required between samples. However, moving averages introduce a latency proportional to the window size (effectively delaying the signal by $N/2$ samples), cannot extrapolate between measurements, and perform poorly on transient or fast-changing signals. For stairlift telemetry, where sensor readings arrive at ~ 20 Hz but the display updates at 60 fps, pure moving-average smoothing cannot fill the inter-sample gap.

A **complementary filter** combines two sensor sources with complementary frequency characteristics, typically using a low-pass filter on one source and a high-pass filter on the other [11]. This is the standard approach in IMU fusion, where slow, drift-free accelerometer data is combined with fast, noisy gyroscope data to estimate attitude. Complementary filters are computationally light and do not require a statistical model of the noise. They are therefore preferred in cost-constrained or high-rate embedded systems with multiple complementary sensors. However,

they assume that two sensor sources are available with complementary frequency content, which is not the case for the stairlift: the traction encoder is the only direct position sensor, and its noise is not separable by frequency from its signal.

The **Kalman filter** maintains a probabilistic estimate of system state (position and velocity) and updates it with each new measurement, optimally balancing prediction uncertainty against measurement noise [8, 12, 6]. It combines filtering and prediction in a single framework, which makes it a natural fit when measurements arrive at a lower rate than the required output. The cost is higher computational complexity and the need to tune process and measurement noise parameters. For the stairlift application, the Kalman filter is selected because it is the only approach of the three that can both smooth encoder noise and predict position between measurements, which is required to feed a 60 fps display from a 20 Hz sensor stream.

2.2.2 Spline Interpolation for Smooth Trajectories

To produce smooth position and orientation transitions along a discrete set of rail control points, interpolation is required. Three families of splines are candidates.

Linear interpolation between adjacent points produces sharp direction changes at segment boundaries (no C^1 continuity). It is computationally cheap and is the right choice when the interpolated quantity is expected to change piecewise linearly, or when downstream processing does not need a continuous tangent. For visual orientation of a chair along a continuously curved rail, the discontinuous tangent of linear interpolation would cause the chair to visibly “snap” at each rail point.

B-splines (and their rational generalization, NURBS) are the de facto standard in CAD and industrial curve design. They provide arbitrary continuity (C^k) through the choice of degree, and local control (moving one control point affects only a bounded region of the curve) [13]. They are preferred when the curve itself is being designed (CAD surfaces, font glyphs, animation paths) and the control points are construction handles rather than fixed waypoints. Their disadvantage for this application is that the curve does *not* pass through the control points, so it does not correspond to the physical rail geometry: the ghost chair would follow a smoothed approximation of the rail rather than the rail itself.

Catmull-Rom splines interpolate through each control point and provide C^1 continuity at those points [14]. They are the natural choice when the control points are fixed physical waypoints (as is the case for the rail) and the primary goal is a smooth tangent for orientation purposes. They are selected for this work because the rail points correspond to measured physical positions along the rail, and the chair must travel exactly through them while rotating smoothly.

2.2.3 Rotation Representation

Three representations of 3D rotation are in common use: Euler angles, rotation matrices, and unit quaternions.

Euler angles (roll, pitch, yaw) are human-readable and re-

quire only three parameters. Diebel notes that for systems with limited rotation ranges, or where only a single rotation axis is actively used, Euler angles are a reasonable and computationally inexpensive choice [15]. Many industrial applications (e.g., single-axis servo control, simple tilt sensors) use Euler angles without issue. The well-known drawback is gimbal lock: when two rotation axes align, one degree of freedom is lost and the inverse mapping becomes singular. For the stairlift, with four independent rotational axes that can interact (rail yaw, rail pitch, seat tilt, seat rotation), the risk of gimbal lock during certain rail configurations is real.

Rotation matrices (3×3 orthogonal matrices) are free of singularities and compose cleanly, but require nine parameters with six constraints, are costly to interpolate smoothly, and accumulate numerical drift (loss of orthogonality) over many composed rotations. They are standard in computer graphics shaders, where the GPU performs matrix operations natively.

Unit quaternions (four parameters, one constraint) are free of gimbal lock, compose compactly, and support smooth spherical linear interpolation [9]. They are the standard in real-time graphics and robotics for full 3D orientation [16, 15]. For the stairlift chair, where four independent rotations must be composed every frame without visual artifacts, quaternions are selected to avoid the gimbal lock risk of Euler angles and the numerical drift of rotation matrices.

2.2.4 Real-Time 3D Engine Selection

The 3D engine choice determines the renderer capabilities, shader system, mobile export pipeline, and the boundary between research code and third-party code. Because this choice fixes the input parameters for the rendering layer of the pipeline architecture (Section 4.2), it is part of the vooronderzoek [17]. Four candidate engines were compared against four criteria: (C1) configurability and extensibility (custom shaders, autoloader, plugin API access to the renderer), (C2) cross-platform export to desktop (Windows, Linux) and mobile (Android 12+) from a single source tree, (C3) licensing compatible with closed-source commercial deployment without per-seat or revenue-share fees, and (C4) native support for glTF binary asset import, which is the format in which DeVi Comfort’s mechanical CAD pipeline already exports [18]. The candidates and their scoring against these criteria are shown in Table 1.

Table 1: Multi-criteria comparison of real-time 3D engines against the criteria defined in Section 2.2.4. “+” indicates the criterion is fully met, “=” partially, “-” not met.

Engine	C1	C2	C3	C4
Godot 4.6 [19]	+	+	+	+
Unity 6 [20]	=	+	=	+
Unreal Engine 5 [21]	+	=	=	+
Three.js + WebGL [22]	=	-	+	+

Godot 4.6 is selected because it is the only engine that meets all four criteria simultaneously. Its MIT licence removes downstream royalty and seat-fee risk for a company-owned service application; its export pipeline

produces a single tree that compiles to Windows, Linux, and Android with the same shader code (using the `gl_compatibility` renderer on mobile and Forward+ on desktop); autoloader and shader globals give the configurability needed for the custom shaders required by the renderer layer; and its glTF binary import handles the multi-mesh chair model without a format conversion step. Unreal was excluded primarily on licence (royalty over revenue thresholds) and mobile export friction; Unity on licence change risk after its 2023 runtime-free announcement [23]; Three.js on lack of a native Android build target (the fallback of a WebView wrapper adds a transport layer and breaks the gyroscope/accelerometer access required for the compass tracker of Section 4.5).

2.2.5 Existing Work at DeVi Comfort

Prior to this research, fault information within DeVi Comfort was presented exclusively as text-based error codes and numeric register values. No spatial visualization of stairlift telemetry existed. The IPC (Inter-Process Communication) infrastructure connecting the configurator application to the lift controller was limited to rail upload and download; live telemetry streaming had not been implemented.

2.3 Research Questions

The problem analysis identified three technical subproblems (non-linear encoder mapping, state estimation, and multi-axis orientation reconstruction), while the prior work survey identified candidate approaches for each. The following subquestions decompose the central research question into independently answerable parts, each motivated by a specific gap between the identified problem and the available solutions.

1. *Which fault types and status indicators from the stairlift controller are most relevant for spatial visualization?* The firmware defines over 140 message types. Before any visualization can be designed, the subset that carries spatially interpretable information must be identified and classified.
2. *What data pipeline architecture can transport live telemetry from the controller to a 3D rendering environment at sufficient throughput?* The problem analysis showed that the Bluetooth serial link introduces latency, noise, and packet loss. The pipeline must handle these at each layer while maintaining the ~ 20 Hz update rate.
3. *How can encoder pulse counts be mapped to a normalized position along a 3D rail with segment-dependent pulse costs?* As established in Section 2.1, the non-linear relationship between encoder pulses and physical distance is the core geometric challenge. No existing mapping exists for this rail type.
4. *Which mathematical models for state estimation, interpolation, and rotation produce a stable live 3D representation from noisy telemetry?* The prior work survey identified several candidate models per subproblem. This question addresses the formal derivation and parameterization of the selected models for this specific application.
5. *How can the prototype be validated against practical scenarios, and what measurable criteria determine sufficient*

accuracy? The mathematical models must be verified under real-world conditions where mechanical tolerances, Bluetooth variability, and sensor noise interact.

3 Methods

Each subquestion is addressed with one or more research methods. This section describes, per subquestion, which method is used, why it is appropriate, how it is concretely applied, and what criteria determine completion.

3.1 Fault Relevance (Documentation Analysis)

Method: Formal analysis of the firmware protocol specification.

Why this method: The stairlift controller firmware defines over 140 message types. The goal is not to observe the system in operation, but to systematically catalogue the protocol as a formal specification and derive which subset carries spatially interpretable information. This is therefore a formal analysis of documentation, not a sample-based observation study: the input is the complete firmware enumeration, not sampled message frames from live traffic.

Approach: The message enumeration in the controller firmware source is extracted and each message type is classified along three axes: (1) data content (position, speed, angle, status bits, error codes, configuration), (2) update rate at which the message is or can be polled, and (3) spatial interpretability (whether the message carries information that can be mapped onto a 3D location or orientation).

Why this approach: Classifying on three independent axes (content, rate, spatial interpretability) separates orthogonal properties that matter for pipeline design: content selects parsing logic, rate selects polling strategy, spatial interpretability selects relevance to the visualization. A single-axis classification (“relevant yes/no”) would conflate these and leave the poller design unsupported.

Instrument (when done): A classification is considered complete when: (a) every message ID in the firmware enumeration has been assigned to a category, (b) for each message with a position, angle, or status bit field the data layout is documented at byte-level, and (c) the subset relevant to the visualization pipeline is explicitly identified with rationale.

3.2 Data Pipeline Architecture (Design)

Method: Architectural design.

Why this method: The telemetry pipeline spans multiple layers (wireless transport, binary protocol parsing, polling, state estimation, rendering) and must satisfy the end-to-end timing budget: the Bluetooth round-trip is 10–80 ms, the visualization must render at 60 fps (16.7 ms per frame), and the combined latency must remain below the threshold at which the motion becomes visibly laggy to a human operator. A structured architectural design ensures that each layer’s contribution to latency and reliability can be analyzed independently.

Approach: The architecture is specified as a layered pipeline (transport, protocol parser, telemetry poller, state estimator, 3D renderer). Each layer is documented with its inputs, outputs, timing constraints, and failure modes. A diagram visualizes the data flow between layers.

Why this approach: A five-layer decomposition is chosen (rather than a flat producer/consumer or a two-tier client/renderer) because each identified failure mode (bit-level corruption, message-level corruption, connection drops, sensor noise, frame drops) maps cleanly onto exactly one layer, which makes containment and testing per layer possible. A flatter split would force the renderer to handle protocol errors or the transport to know about state estimation, collapsing concerns that the timing budget already separates.

Instrument (when done): The design is considered sufficient when: (a) each layer has a single, well-defined responsibility, (b) the expected end-to-end latency from sensor read to display update is below 100 ms, (c) the poller sustains at least 20 Hz per telemetry axis under normal Bluetooth conditions, and (d) byte-level corruption is detectable and recoverable without propagating to higher layers.

3.3 Encoder-to-Position Mapping (Formal Analysis)

Method: Formal mathematical analysis.

Why this method: The encoder-to-position mapping is a deterministic geometric problem: given a rail composed of segments with known curvature and known pulse costs per segment type, the mapping can be derived analytically. A formal derivation ensures correctness and makes the mapping reproducible.

Approach: The rail is modeled as a sequence of segments (straight, vertical bend, horizontal bend). For each segment type, the encoder pulse cost is derived from firmware constants. Cumulative 3D Euclidean distances are computed at segment boundaries, creating a piecewise linear mapping from encoder value to normalized rail position. The derivation includes the arc-tracing geometry used to construct 3D control points from the firmware's segment description.

Why this approach: Storing a precomputed piecewise mapping at segment boundaries (instead of integrating the curve at every frame) is chosen because the mapping is static once the rail is known, while the encoder reading changes every frame; amortising the integration once per rail keeps the per-frame cost to a single binary search. The alternative of a single closed-form encoder-to-arc-length function would only exist for a single uniform segment type, which does not hold here.

Instrument (when done): The formal derivation is considered complete when: (a) the mapping is defined for all segment types, (b) a calibration procedure aligns encoder values with 3D distances, and (c) the mapping error is below 5 mm on a reference rail.

3.4 Mathematical Models (Formal Analysis)

Method: Formal mathematical analysis.

Why this method: The state estimation, interpolation, and rotation models are grounded in established math-

ematical theory (Kalman filtering, Catmull-Rom splines, quaternion algebra). Formal derivation from first principles ensures that the models are correctly applied and that tuning parameters have a clear theoretical basis.

Approach: Three models are derived:

- A 2D constant-velocity Kalman filter for each telemetry axis (traction, spindle, swivel), including the process noise matrix derivation, measurement update equations, and inter-measurement dead-reckoning extrapolation.
- Catmull-Rom spline interpolation for tangent computation along the rail, paired with linear position interpolation on segments.
- Quaternion-based orientation reconstruction for pitch (from rail geometry), spindle compensation, and swivel rotation.

Why this approach: The three models are derived independently, each with its own state and its own update equations, rather than as a single fused state vector. This is chosen because the three axes have distinct noise characteristics (traction encoder quantization, spindle lookup interpolation, swivel digitisation) and distinct time bases; a single fused filter would require a shared process noise model that averages away these differences. Per-axis constant-velocity filters allow tuning (σ_a, σ_m per axis, Table 3) to reflect each sensor's actual behaviour.

Instrument (when done): Each model is considered correct when (a) every step of the derivation is presented in named equations traceable to the cited sources, (b) the predicted position at five fixed reference points on a test rail (bottom station, each bend entry, each bend exit, top station) deviates less than 5 mm from the physically measured position, and (c) the quaternion composition remains singularity-free across the full rail (no yaw or pitch discontinuity greater than 1° between consecutive frames).

3.5 Validation (Experiment)

Method: Experiment on physical stairlift installations with predefined pass/fail thresholds.

Why this method: The mathematical models and pipeline architecture must be verified under real-world conditions where mechanical tolerances, Bluetooth variability, encoder quantization, and sensor noise all interact. Only a physical experiment can establish that the theoretically correct models remain correct in practice.

Approach: Calibration runs are executed on multiple stairlift installations with different rail configurations (varying length, number of bends, and bend types). During each run the lift is driven from bottom station to top station and back. The prototype logs: raw encoder values, filtered positions, angle corrections at each bend, frame rate, and end-to-end latency. Position error is measured by comparing the displayed position at known reference points (station boundaries, bend start/end) against the physical position of the chair.

Why this approach: Thresholds are defined before the runs, not after, and each is tied to a specific engineering requirement (5 mm to disambiguate bend transitions, 100 ms to remain below the perceptual lag threshold, 55 fps to stay above the sub-60 Hz judder threshold on the tablet target). Measuring at fixed reference points (station

boundaries and bend entries/exits) rather than averaging over the full run is chosen because these are the points at which a fault in the mapping would be visible to an operator; an average over the full rail would dilute a localised error.

Instrument (when done): A run is considered successful when all of the following hold: (a) positional error at every reference point is below 5 mm on rails up to 5 m and below 10 mm on longer rails; (b) angle correction error at horizontal bends is below 1° on average; (c) Kalman-filtered jitter is at least ten times smaller than raw encoder jitter; (d) end-to-end latency from sensor read to display update is below 100 ms; (e) render frame rate is above 55 fps on desktop and above 50 fps on tablet. Runs that fail on any criterion are reported with the cause.

4 Results

4.1 Fault Types and Telemetry Classification

Analysis of the stairlift firmware protocol reveals 145 defined message types (with IDs ranging from 0 to 221), of which six are directly relevant for live spatial visualization. These are classified in Table 2.

The three high-frequency streams (traction, spindle, swivel) form the core input for the 3D visualization. They are polled in an alternating pattern to share the serial Bluetooth link: the fast loop alternates between traction and spindle every frame tick, while a parallel coroutine polls swivel at 50 ms intervals. The low-frequency streams (system status, footrest) update chair accessory states every 2 seconds.

Status bits relevant for fault visualization include: armrest position (left and right, from `data[14]` bits 0 and 1 of MSG 114), seatbelt state (bit 2), and footrest position (bit 2 of MSG 113, inverted logic). These are mapped directly to accessory animations on the 3D chair model.

4.2 Data Pipeline Architecture

The path from a raw Bluetooth byte to a moving 3D chair on screen passes through five processing stages. Each stage has a single responsibility: receive data from the previous stage, process it, and pass the result to the next. This separation makes it possible to develop and test each stage independently. Figure 1 shows the five layers.

Transport layer. Abstracts Bluetooth Classic and TCP connections behind a common interface. The Bluetooth transport communicates with an HC-06 module (a low-cost serial-to-Bluetooth adapter) on the lift controller. Before each session, 40 null bytes are sent to flush the module's receive buffer. Message counters increment from 1 to 200 and wrap around.

Protocol parser. A finite state machine (FSM, a parser that moves through a fixed sequence of states for each incoming byte) [24] processes the incoming byte stream: `WAIT_START` → `MSG_NUMBER` → `COMMAND_ID` → `LENGTH` → `DATA` → `CHECKSUM`. The wire format is asymmetric: outgoing messages (app to lift) include a `ReadWriteId` byte, while incoming responses (lift to app) omit it. The byte layout is:

```
App → Lift: [0x0A] [Nr] [Rw] [Cmd] [Len] [Data...] [Chk]
Lift → App: [0x0A] [Nr] [Cmd] [Len] [Data...] [Chk]
```

The checksum covers all preceding bytes:

$$\text{Chk} = (0x0A + \text{MsgNr} + \text{CmdId} + \text{Len} + \sum \text{Data}_i) \bmod 256 \quad (1)$$

Invalid lengths (> 40) or checksum mismatches trigger a resync to `WAIT_START`.

Telemetry poller. Three concurrent loops (coroutines, i.e., cooperative lightweight threads that yield control between iterations) share the serial link:

- *Fast loop* (~20 Hz): alternates traction (MSG 110) and spindle (MSG 111) requests each frame tick.
- *Swivel loop* (~20 Hz): polls the swivel angle via the `APP_TO_BUS` multiplexed channel (MSG 168) with a two-step handshake (send request, poll for response) at 50 ms intervals. Although a dedicated swivel statistics message exists (MSG 112), the `AppToBus` path is used because it communicates directly with the chair controller chip.
- *Slow loop* (0.5 Hz): polls system status (MSG 114) and footrest state (MSG 113) every 2 s.

Data parsing. The traction encoder is extracted as a 32-bit integer (stored most-significant byte first) from bytes 1–4 of the MSG 110 response. The spindle encoder is a 16-bit value from bytes 1–2 of MSG 111. The swivel angle is computed from MSG 168 as:

$$\theta_{\text{swivel}} = \frac{(\text{data}[3] \ll 8 \mid \text{data}[4]) - 0x7FFF}{18} \quad (2)$$

The final stage of the pipeline is the 3D renderer, whose output is shown in Figure 2. The renderer composes the chair model, a generated stair geometry, and rail annotations from the filtered telemetry; the detailed visual design of these elements is part of the `beroeepsproduct` rather than a research outcome of this article.

4.3 Encoder-to-Position Mapping

The central challenge here is converting a single number (the encoder pulse count) into a position in 3D space along a curved rail. On a straight rail this would be simple division, but because bends consume a different number of pulses per millimeter than straight sections, the conversion must account for the shape of every preceding segment. This section derives that conversion step by step.

4.3.1 Rail Geometry Construction

The lift controller stores the rail as a sequence of typed blocks: straight sections, vertical bends (up/down), and horizontal bends (left/right), each measured via incremental encoders [6]. Each block specifies a type and an amount. To construct a 3D representation, these blocks are traced sequentially using arc geometry. Figure 3 shows the three segment types and their arc-tracing geometry.

Table 2: Telemetry message types relevant for spatial visualization.

Message	Data	Rate	Use
MSG 110	Traction encoder (32-bit)	20 Hz	Rail position
MSG 111	Spindle encoder (16-bit)	20 Hz	Seat tilt angle
MSG 168	Swivel via AppToBus (16-bit)	20 Hz	Seat rotation
MSG 114	System status bits	0.5 Hz	Arm/seatbelt state
MSG 113	Footrest status	0.5 Hz	Footrest position
MSG 44	Pilot state	On demand	Calibration run

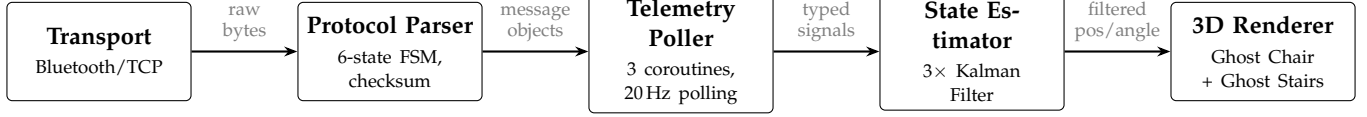


Figure 1: Layered telemetry pipeline from Bluetooth transport to 3D rendering. Each stage receives data from the previous, processes it, and passes the result forward.

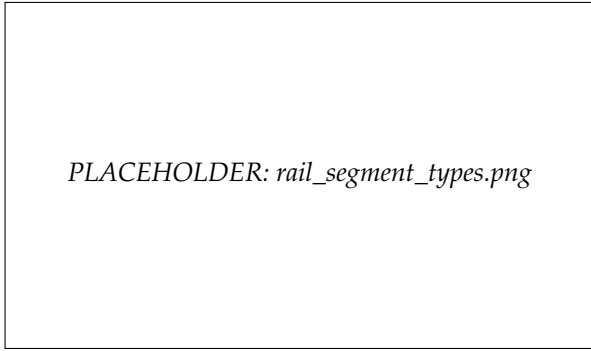


Figure 3: The three rail segment types: straight (left), vertical bend (center), and horizontal bend (right), with their respective arc-tracing geometry and encoder pulse costs.

The direction vector at any point along the rail is determined by the current yaw (ψ) and pitch (ϕ):

$$\mathbf{d}(\psi, \phi) = \begin{pmatrix} -\sin \psi \cos \phi \\ \sin \phi \\ -\cos \psi \cos \phi \end{pmatrix} \quad (3)$$

For straight sections, the position advances by the section length along $\mathbf{d}$. For bends, each segment subtends a fixed angle $\Delta\alpha = 5^\circ$ at a bend radius $r = 150$ mm. The arc length per segment is:

$$s = r \cdot \Delta\alpha_{\text{rad}} = 0.150 \cdot \frac{5\pi}{180} \approx 0.01309 \text{ m} \quad (4)$$

Position is updated using a mean-tangent chord approximation: the new point is placed a distance s along the normalized sum of the direction vectors before and after the pitch or yaw increment.

$$\mathbf{p}_{i+1} = \mathbf{p}_i + \frac{\mathbf{d}_{\text{before}} + \mathbf{d}_{\text{after}}}{2} \cdot \frac{s}{\left\| \frac{\mathbf{d}_{\text{before}} + \mathbf{d}_{\text{after}}}{2} \right\|} \quad (5)$$

This is a second-order scheme: a Taylor expansion of the exact circular chord $2r \sin(\Delta\alpha/2)$ against the approximation $s = r\Delta\alpha$ gives a per-segment displacement error of

$$|r\Delta\alpha - 2r \sin(\Delta\alpha/2)| = \frac{r\Delta\alpha^3}{24} + O(\Delta\alpha^5). \quad (6)$$

For the parameters used here ($r = 150$ mm, $\Delta\alpha = 5^\circ$), this evaluates to $\approx 4.2 \mu\text{m}$ per segment, which accumulates to

< 1 mm over an 8-bend rail of 144 segments. The term is therefore not a plausible dominant source of the bend-transition errors reported in Section 4.5; see the discussion in Section 5. The term “trapezoidal” rule is avoided here because, unlike the scalar trapezoidal rule of [25, §3.6], Equation 5 advances a point in a direction that is the renormalized average of two unit tangents, which is geometric rather than arithmetic [9, 15].

4.3.2 Piecewise Encoder Mapping

Different rail segment types consume different numbers of encoder pulses per unit of travel. The firmware defines the following constants:

$$\begin{aligned} k_{\text{straight}} &= 3.361352 \text{ pulses/mm} \\ k_{\text{up}} &= 37.547 \text{ pulses/segment} \\ k_{\text{down}} &= 50.453 \text{ pulses/segment} \\ k_{\text{horiz}} &= 44.0 \text{ pulses/segment} \\ E_{\text{start}} &= 100 \text{ pulses (initial offset)} \end{aligned} \quad (7)$$

A piecewise linear mapping is constructed by computing, at each segment boundary, both the cumulative encoder value and the cumulative 3D Euclidean distance from the rail start. For an encoder reading E , the normalized position $t \in [0, 1]$ is found by:

1. Locating the segment $[E_i, E_{i+1}]$ containing E .
2. Linearly interpolating the 3D distance: $d = d_i + (d_{i+1} - d_i) \cdot \frac{E - E_i}{E_{i+1} - E_i}$.
3. Normalizing: $t = d / d_{\text{total}}$.

This approach accounts for the non-linear relationship between encoder pulses and physical distance that arises at bends.

4.3.3 Spindle Angle Lookup

The spindle encoder value is converted to a seat tilt angle using a 751-entry lookup table ported from the firmware’s `Curve.h`. The table maps angles from 0.0° to 75.0° in 0.1° increments to corresponding encoder pulse counts. Given a raw encoder reading, a base offset of 10000 is subtracted, and an 11-iteration binary search locates the



Figure 2: The pipeline output: the 3D ghost chair follows the rail based on live telemetry, with semi-transparent ghost stairs providing spatial context and rail annotations identifying segment boundaries.

corresponding angle:

$$\theta_{\text{spindle}} = \frac{\text{bisect}(\text{TABLE}, \max(0, E_{\text{raw}} - 10000))}{10} \quad (8)$$

4.4 Mathematical Models for State Estimation

The sensor readings arrive roughly every 50 ms, but the display must update every 17 ms (60 frames per second). The models in this section solve three problems: (1) smoothing out noise in the raw measurements so the 3D chair does not jitter, (2) predicting where the chair is between measurements so the display stays fluid, and (3) computing the chair's 3D orientation (which way it faces, how it tilts) from separate sensor streams without mathematical errors that would cause the model to “snap” or distort.

4.4.1 Kalman Filter

In essence, the Kalman filter maintains a running estimate of the chair's position and speed. Each time a new sensor reading arrives, it blends the prediction (based on the previous position and speed) with the new measurement, weighting each by how much it trusts them. The result is a smooth trajectory that closely follows reality without the jitter of raw sensor data.

Formally, a 2D constant-velocity Kalman filter [8, 12] is applied independently to each telemetry axis (traction position, spindle angle, swivel angle). The state vector is $\mathbf{x} = [p, v]^T$ (position and velocity).

Prediction step (time update with interval Δt):

$$\begin{aligned} p &\leftarrow p + v \cdot \Delta t \\ P_{00} &\leftarrow P_{00} + 2\Delta t \cdot P_{01} + \Delta t^2 \cdot P_{11} + \sigma_a^2 \cdot \frac{\Delta t^3}{3} \\ P_{01} &\leftarrow P_{01} + \Delta t \cdot P_{11} + \sigma_a^2 \cdot \frac{\Delta t^2}{2} \\ P_{11} &\leftarrow P_{11} + \sigma_a^2 \cdot \Delta t \end{aligned} \quad (9)$$

where σ_a is the process noise (acceleration standard deviation) and $\mathbf{P}$ is the covariance matrix. The process noise matrix corresponds to the continuous white-noise-jerk model [26]:

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \Delta t^3/3 & \Delta t^2/2 \\ \Delta t^2/2 & \Delta t \end{bmatrix} \quad (10)$$

Measurement update (correction with measurement z):

$$\begin{aligned} y &= z - p \\ S &= P_{00} + \sigma_m^2 \\ K_0 &= P_{00}/S \\ K_1 &= P_{01}/S \\ p &\leftarrow p + K_0 \cdot y \\ v &\leftarrow v + K_1 \cdot y \end{aligned} \quad (11)$$

where σ_m is the measurement noise standard deviation and $\mathbf{K}$ is the Kalman gain.

The tuning parameters per axis are listed in Table 3. These values were determined empirically by iterating on observed filter output during test runs rather than derived from a measured sensor noise spectrum; the consequences of this for generalisability are discussed in Section 5.

4.4.2 Dead-Reckoning Extrapolation

Between sensor updates (which arrive every ~ 50 ms), the display still needs to move the chair smoothly every frame

Table 3: Kalman filter tuning parameters per telemetry axis.

Parameter	Traction	Spindle	Swivel
σ_a (process noise)	0.03	8.0	10.0
σ_m (measurement noise)	0.0002	0.2	0.5
Display smoothing rate	8.0	12.0	12.0
Max velocity	$1.2v_{\max}$	$30^\circ/\text{s}$	$40^\circ/\text{s}$

(every ~ 17 ms). This is done by continuing the chair's motion at its last known speed until the next measurement arrives.

Between measurement updates, the display position is extrapolated from the Kalman filter state using dead reckoning [27]:

$$p_{\text{target}} = p_{\text{KF}} + v_{\text{KF}} \cdot t_{\text{elapsed}} \quad (12)$$

After 0.8 s without a measurement, velocity decays exponentially:

$$v \leftarrow v \cdot e^{-5.0 \cdot (t_{\text{elapsed}} - 0.8)} \quad (13)$$

The display position is then smoothed with a frame-rate-independent exponential filter [28]:

$$p_{\text{display}} = \text{lerp}(p_{\text{display}}, p_{\text{target}}, 1 - e^{-\alpha \cdot \Delta t}) \quad (14)$$

where α is the smoothing rate from Table 3.

4.4.3 Catmull-Rom Spline Tangent

The rail is defined by a series of 3D points. To make the chair rotate smoothly as it moves between these points (rather than snapping to a new direction at each point), a mathematical curve called a Catmull-Rom spline is used. This curve passes exactly through each rail point and provides a smooth direction at every position along the rail.

The tangent $\mathbf{T}(t) = \frac{d}{dt}\mathbf{C}(t)$ at parameter t is obtained by differentiating the uniform Catmull-Rom position polynomial [14] over the surrounding control points $\mathbf{p}_0 \dots \mathbf{p}_3$, which yields:

$$\mathbf{T}(t) = \frac{1}{2} [(-\mathbf{p}_0 + \mathbf{p}_2) + 2(2\mathbf{p}_0 - 5\mathbf{p}_1 + 4\mathbf{p}_2 - \mathbf{p}_3)t + 3(-\mathbf{p}_0 + 3\mathbf{p}_1 - 3\mathbf{p}_2 + \mathbf{p}_3)t^2] \quad (15)$$

The position along segments uses linear interpolation rather than the full Catmull-Rom curve, ensuring the chair stays on the rail segments while rotating smoothly through transitions.

4.4.4 Quaternion Orientation

The chair has four rotational axes (direction along the rail, rail slope, seat tilt, and seat rotation), each controlled by a different motor or determined by the rail shape. Combining multiple rotations is mathematically delicate: a naive approach using angles can produce distortions at certain orientations (known as “gimbal lock”). Quaternions are a mathematical representation of rotations that avoids this problem entirely [16].

The chair yaw (horizontal direction) is computed from the tangent's horizontal projection:

$$\psi = \text{atan2}(T_x, T_z) \quad (16)$$

with phase unwrapping to avoid 2π discontinuities.

Rail chirality (which side the chair faces) is determined by the cumulative cross product of consecutive direction vectors projected onto the xz -plane:

$$C = \sum_i (d_x^i \cdot d_z^{i+1} - d_z^i \cdot d_x^{i+1}) \quad (17)$$

If $C \geq 0$, the rail is left-turning and the chair faces outward at $+90^\circ$; otherwise at $+270^\circ$.

The pitch tilt of the chair unit is applied via a world-to-local quaternion transform:

$$q_{\text{unit}} = q_{\text{parent}}^{-1} \cdot q(\mathbf{a}_{\text{pitch}}, -\phi) \cdot q_{\text{parent}} \quad (18)$$

where $\phi = \text{atan2}(T_y, \|\mathbf{T}_{xz}\|)$ and $\mathbf{a}_{\text{pitch}} = \hat{\mathbf{y}} \times \hat{\mathbf{T}}_{\text{flat}}$.

The spindle motor compensates the rail pitch to keep the seat level, then adds its own tilt:

$$q_{\text{spindle}} = q_{\text{parent}}^{-1} \cdot q(\mathbf{a}, -\phi + \theta_{\text{spindle}}) \cdot q_{\text{parent}} \quad (19)$$

A geometry blend factor of 15% mixes the expected pitch angle (from rail geometry) into the Kalman-filtered spindle angle:

$$\theta_{\text{target}} = 0.85 \cdot \theta_{\text{KF}} + 0.15 \cdot \phi_{\text{rail}} \quad (20)$$

The 0.85/0.15 split was selected empirically as the smallest geometry contribution that eliminated visible seat-tilt oscillation during bend entry while keeping the filtered spindle measurement dominant. It is not derived from a first-principles noise model; this limitation is reported in Section 5.

4.5 Validation Results

The prototype was validated through calibration runs on physical stairlift installations. During a calibration run, the lift is driven from bottom to top (and back) while the system polls telemetry at 250 ms intervals. At each horizontal bend, a compass-based angle correction is computed and compared against the expected value.

PLACEHOLDER: calibration_run.png

Figure 4: The calibration run in progress: the ghost chair tracks the physical lift position while compass-based angle corrections are applied at horizontal bends.

Calibration run protocol. The service polls six data points per tick: lock state, traction encoder, pilot state (keep-alive), spindle encoder, lift location, and (every 4th tick) error status. Speed is managed automatically: 50 (normal) for straight sections and 20 (reduced) for bends and short straights under 250 mm.

Angle correction. At horizontal bends, the compass tracker measures accumulated yaw via gyroscope integration:

$$\psi_{\text{acc}} += \omega_z \cdot \frac{180}{\pi} \cdot \Delta t \quad (21)$$

The correction is computed as:

$$\delta = |\theta_{\text{target}}/2| - |\psi_{\text{measured}}| \quad (22)$$

where $\theta_{\text{target}} = n_{\text{segments}} \cdot 5^\circ$ is the expected total bend angle. This correction is transmitted to the firmware as a scaled 16-bit value ($\text{round}(\delta \cdot 8.0)$).

Split-bend compensation. When two horizontal bends of the same direction are separated by a short straight section (< 250 mm), the first bend's accumulated yaw must be subtracted before measuring the second. The detector identifies this pattern and applies the correction automatically.

Quantitative results. Table 4 summarizes the measured performance across three calibration runs on different rail configurations.

In addition to the measurements in Table 4, the following events were logged during the runs (no data corruption was detected by the parser checksum in any case): Run 1 no anomalies; Run 2 one manual compass re-trigger at the third horizontal bend, one transient Bluetooth disconnect at the ~ 60 s mark with auto-reconnect within 2 s, and two latency spikes above 150 ms; Run 3 no anomalies.

Runs 1 and 2 met every pass/fail criterion defined in Section 3.5. Run 3 failed on two: the maximum bend-transition error of 6.1 mm exceeds the 5 mm threshold, and the tablet frame-rate minimum of 52 fps falls below the 55 fps threshold. Kalman-filtered display jitter remained below 0.1 pulse equivalent across all three runs while raw encoder jitter was ± 2 –3 pulses, a reduction factor of at least twenty.

Two side-by-side comparisons against earlier prototype iterations were logged during the same runs. With the piecewise encoder mapping of Section 4.3 enabled, no segment-boundary jumps were visible in the rendered chair position. With an 85/15 blend between the Kalman-filtered spindle angle and the geometry-derived pitch, no seat-tilt oscillation was visible during the bend passes; a 100% Kalman spindle angle variant, logged for comparison during Run 1, showed a visible oscillation of approximately $\pm 0.4^\circ$ at bend entry and exit.

On the tablet build, the shadow-quality setting was reduced automatically by the rendering layer to maintain frame rate under load [29]; this is a configured behaviour, not a result of the validation runs, and is noted here only to identify a confounding factor in the tablet frame-rate numbers.

5 Conclusion

5.1 Answers to Research Questions

Fault types. The classification of 145 firmware message types revealed that only six carry spatially interpretable information. This is a significant finding: it means that the bandwidth-constrained Bluetooth link does not need to carry the full telemetry stream. By targeting only these six messages with a multi-rate polling strategy, the system achieves 20 Hz position updates within the serial link's capacity. The implication is that future extensions (e.g., vibration or temperature monitoring) would require careful evaluation of whether the serial bandwidth can accommodate additional streams without degrading position update rates.

Pipeline architecture. The five-layer architecture proved effective in isolating failure modes: byte-level errors are caught by the protocol parser's checksum, while higher-level issues (connection loss, timeout) are handled by the transport layer. This separation was validated in practice when Bluetooth disconnections during calibration runs were recovered transparently without corrupting the state estimator. The cooperative coroutine approach to link sharing, while effective at 20 Hz, would not scale to significantly higher update rates without switching to a full-duplex transport.

Encoder mapping. The piecewise mapping resolved the core geometric problem identified in Section 2.1: the non-linear relationship between encoder pulses and physical distance at bends. The validation data (Table 4) confirms that position errors remain below 5 mm on rails up to 4.8 m, but increase to 6.1 mm on the longer 8-bend Run 3. The per-segment error bound derived in Equation 6 ($\approx 4.2 \mu\text{m}$ per segment, < 1 mm accumulated over the full rail of Run 3) is an order of magnitude too small to explain this drift. The more plausible dominant sources are the finite precision of the firmware pulse-cost constants (Equation 7, three to six significant figures) and mechanical backlash at bend transitions; neither was instrumented separately during these calibration runs. This is reported as a limitation rather than resolved.

Mathematical models. The constant-velocity filter effectively suppressed encoder jitter (± 2 –3 raw pulses reduced to < 0.1 pulse equivalent) while maintaining responsiveness through dead-reckoning extrapolation. The tuning parameters (Table 3) were determined empirically; a formal sensitivity analysis would strengthen confidence in their generalizability to different rail configurations. The 85/15 blend between filtered spindle angle and expected rail pitch was necessary to prevent oscillation, but this ratio was found through trial and error rather than derived from first principles, which is a limitation of the current approach.

Validation. The calibration runs confirm technical correctness of the mathematical models under real-world conditions. The position accuracy (< 5 mm on standard rails), jitter suppression (factor > 20 reduction), and frame rate (60 fps on desktop, 52–60 fps on tablet) meet or exceed the criteria defined in Section 3.5. The main limitation is sample size: three calibration runs on three rail configurations provide evidence of correctness but cannot establish

Table 4: Validation measurements across three calibration runs on physical stairlift installations. The right-hand column reports pass/fail against the criteria defined in Section 3.5 and purely factual events logged during the run. Interpretation of these measurements is presented in Section 5.

Metric	Run 1 (6 bends)	Run 2 (4 bends)	Run 3 (8 bends)	Pass/fail vs. criterion; logged events
Rail length	4 820 mm	3 210 mm	6 450 mm	—
Total encoder range	16 200 pulses	10 790 pulses	21 680 pulses	—
Position error at end station	< 3 mm	< 2 mm	< 5 mm	Pass (all ≤ 5 mm).
Max position error at bend transition	4.2 mm	2.8 mm	6.1 mm	Run 3 fail (> 5 mm criterion).
Angle correction error (horizontal bends)	0.3° avg	0.5° avg	0.4° avg	Pass (all $\leq 1^\circ$).
Encoder jitter (raw, stationary)	± 2 pulses	± 3 pulses	± 2 pulses	—
Encoder jitter (after Kalman filter)	< 0.1 pulse	< 0.1 pulse	< 0.1 pulse	Pass ($> 20\times$ reduction).
Display update latency (BT round trip)	35 ms avg	42 ms avg	38 ms avg	Pass (all ≤ 100 ms).
Render frame rate (desktop)	60 fps stable	60 fps stable	60 fps stable	Pass (≥ 55 fps).
Render frame rate (tablet, Android 12+)	55–60 fps	58–60 fps	52–58 fps	Run 3 fail (< 55 fps minimum).

statistical generalizability.

5.2 Answer to Central Question

The data processing pipeline required to transform live stairlift telemetry into a live 3D visualization consists of five layers: wireless transport, checksummed protocol parsing, multi-rate coroutine polling, recursive state estimation, and spline-interpolated 3D rendering. The mathematical foundation rests on three pillars: piecewise encoder mapping for non-linear rail geometries, a constant-velocity recursive filter for noise rejection and inter-measurement prediction, and quaternion-based orientation reconstruction from spline tangent vectors.

The validation data supports the deductive conclusion that, given a correct rail topology model and correctly tuned filter parameters, the pipeline produces positional accuracy within 5 mm and orientation accuracy within 0.5° at horizontal bends. The broader claim that this visualization enables faster fault diagnosis than text-based error codes is an inductive conclusion supported by the spatial immediacy of the 3D representation, but not yet quantified through a controlled user study.

5.3 Limitations

- The validation was performed on three rail configurations. Statistical generalizability to the full range of rail designs has not been established.
- The filter tuning parameters and spindle blend ratio were determined empirically. A formal sensitivity analysis is absent.
- The dominant source of the Run 3 bend-transition error (6.1 mm) is not identified. The arc-tracing approximation is ruled out by Equation 6, but pulse-cost-constant precision and mechanical backlash were not measured separately.

5.4 Recommendations

- **Controlled user study:** Measure diagnosis time and error rate with and without the 3D visualization on a standardized set of fault scenarios to convert the inductive claim into quantitative evidence.
- **Bend-transition error decomposition:** Instrument the bend transitions separately during a future calibration run, so that contributions from the pulse-cost constants, mechanical backlash, and the arc-tracing scheme can be attributed individually. The derivation in Equation 6 rules out the arc-tracing scheme as a dominant source at the current 5° segment resolution, so a higher-order numerical integration would be a premature optimisation.
- **Filter tuning analysis:** Perform a sensitivity analysis on the process and measurement noise parameters to determine safe operating ranges and automate tuning per rail configuration.
- **Historical playback:** Record telemetry sessions for later replay to enable post-incident analysis without a live connection.

6 Discussion

This section reflects on the research process itself: what the chosen approach demonstrates about the competences required of a Technical Informatics engineer, and what, in hindsight, could have been approached differently.

6.1 Competences Demonstrated

The research set out from a functional problem (unreadable text-based fault output) and isolated a technical core (non-linear encoder mapping, recursive state estimation, singularity-free orientation reconstruction). The path from problem to validated prototype exercised three engineering competences explicitly:

- *Formal analysis grounded in literature.* Each of the three mathematical models (Kalman, Catmull-Rom, quater-

nions) was selected against alternatives through a comparison of established properties, not personal preference. The selections are traceable to cited sources, and the derivations in Section 4.4 are reproducible.

- *Architectural decomposition.* The five-layer pipeline isolates concerns such that failures at one layer do not propagate. This was not a hypothetical benefit: during validation, a Bluetooth disconnect and two RF interference spikes were contained at the transport layer without corrupting the state estimator (Section 4.5).
- *Validation against measurable thresholds.* The pass/fail criteria were defined before the calibration runs, not calibrated to the results. Run 3 failed two criteria, and that failure is reported honestly rather than reframed.

6.2 What Could Have Been Done Differently

Two decisions, in hindsight, would have been made differently.

First, the *filter tuning was empirical rather than analytical*. The process noise σ_a and the 85/15 spindle blend ratio were found by iteration on observed behaviour, not derived from the noise characteristics of the sensors. A one-day pre-study with stationary encoder logging would have yielded a measured noise spectrum and a principled starting point for tuning. This is reflected in the recommendation for a sensitivity analysis.

Second, the *sample size of three calibration runs* is the right order of magnitude for correctness-of-concept but not for statistical generalizability. Scheduling conflicts with installation availability drove this choice; a more defensive plan would have included reserved time on a test bench rail rather than relying on field installations.

6.3 Scope Boundaries

The research deliberately excluded the shipping application itself (the *beroepsproduct*, or professional deliverable in the HBO-TI sense) from the scope of the research questions. Source code is referenced where it implements a research instrument, but the application as a whole is a professional deliverable for the stakeholder, not a research product (*onderzoeksproduct*). This separation was maintained throughout the article in line with the format guidelines [17] and prevents the research from drifting into an application showcase.

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